

GReinSS Tutorial — Speaker Notes

NCI Spring School on Algorithmic Cancer Biology

One block per slide. → NOTEBOOK slides are the live-demo hand-offs.

1. GReinSS

Hi everyone. Today: a hands-on tutorial on GReinSS — a method for a problem that shows up all over algorithmic cancer biology: inferring hidden combinatorial states from noisy, indirect measurements. We'll cover the idea, the one theorem that makes it work, and then train it live on your laptop. Goal: you leave able to apply it to your own problem.

2. The recurring problem in computational biology

The unifying pattern: a hidden discrete structure S , indirect observation X , and a KNOWN or partially-known likelihood $\Pr(X|S)$. Trees, CNA sets, isoforms — all fit. This is “self-supervised”: the physics/biology of measurement is known; the state is not.

3. Two things we want

Panel a: hidden distribution over states S^* , each emits an X . We model $\Pr(S|\theta)$. Two problems: (1) LEARN θ from all observations jointly — the shared model couples them; (2) INFER the best state per observation. Everything today serves these two.

4. Why the usual tools struggle

EM: exact expectation needs summing over all states — only works for special structure (HMMs). VAE: great generative models, but the latent lives in a made-up $\mathbb{R}^d$, not your isoform space. Local search: per-observation, no sharing of statistical strength. Naive PG / GFlowNet are the closest cousins to what we do — and we'll see exactly why they fail.

5. The GReinSS idea

(no notes)

6. Generate states as trajectories

This is the RL move: represent a big discrete object as a path of small decisions. The policy is a neural net that at each step picks the next action. Any

structure you can grow incrementally fits. We never enumerate S — we sample trajectories and average.

7. The one equation that matters

This is the whole method in one line. The numerator $\Pr(X_i|\tau)$ is the usual “how well does this trajectory explain observation i ”. The DENOMINATOR $\Pr(X_i|\theta)$ is the current model’s total probability of that observation — it rescales each observation’s contribution. Gradient is taken ONLY through $\log \Pr(\tau|\theta)$; the reward is treated as a constant each step, then recomputed after the update. That’s the “dynamic” part.

8. Why the denominator? (intuition)

Key teaching moment. Without rescaling, τ_1 has the highest raw reward, so naive PG puts ALL mass on it — but then X_2 has zero probability and the joint likelihood is ZERO. With the denominator, as soon as τ_1 gets probability its reward drops (it’s dividing by its own success), so the policy is pushed to also cover X_2 . Equilibrium = the likelihood optimum. Note τ_3 dies: it’s dominated by τ_2 for explaining X_2 . The method finds the RIGHT support.

9. The reward is dynamic — that’s the whole trick

Payoff of the title. The numerator is per-trajectory fit; the denominator is the current model’s total probability of X_i — it shifts as θ learns, so we can’t precompute it. Each iteration we sample trajectories and average $\Pr(X_i|\tau)$ to estimate it. As a state gets covered, its denominator grows and its reward shrinks — automatic load balancing.

10. The training loop

Emphasize the API surface: the user supplies (a) a generator and (b) $\Pr(X|S)$. That’s it. The reward machinery, sampling, and gradient are provided. This is exactly what the notebook will show — you’ll write those two functions and call `train()`.

11. Off-policy learning (when on-policy is too slow)

Practical must-have for hard problems. Instead of blindly sampling from the policy, we tilt sampling toward states that actually fit each observation — provably the best proposal. In our biology applications this is where domain knowledge enters: a fast classical method proposes candidate states, and GReinSS refines the distribution over them. Keep this slide brief unless the audience asks.

12. The scoreboard — who actually maximizes the likelihood?

The positive flip of the opening “why tools struggle” table. Same objective across the board; only GReinSS provably maximizes the joint data likelihood. Naive PG collapses to one state; GFlowNets match rewards, not likelihood; EM-based generative models approximate via point estimates; local search has no shared model. This is the one-slide “why us” scoreboard.

13. → NOTEBOOK · Demo 1: Set reconstruction

SWITCH TO JUPYTER. Walk through: load observations → define $\Pr(X|S)$ (one line) → build generator net → train ~200 epochs live (watch the likelihood curve rise) → infer states → compare to naive thresholding. The punchline: GReinSS denoises using structure shared across observations, beating per-pixel rounding.

14. Results — simulations

Two combinatorial state types, same method. Left: graphs — GReinSS dominates especially when observations are information-poor (few walks). Right: sets — GReinSS is the only method that scales to large universes. GEM-based methods (VAE/autoregressive/diffusion) plateau; the closest RL cousins (naive PG, GFlowNet) fail. The reward rescaling is the difference.

15. → NOTEBOOK · Demo 2: Graph inference (pre-trained)

SWITCH TO JUPYTER (second section). Heavier model, so we ship a pre-trained checkpoint. Show: load model → simpleInference → compare predicted adjacency to the ground-truth graph we saved during pre-training → report F1 and visualize one graph. This mirrors the paper’s Fig on graph inference but on your own generated instance with known truth.

16. Application — RNA isoforms beat RSEM

The payoff for this audience. Isoform quantification is a textbook latent-variable problem: short reads are indirect observations of full-length transcripts. RSEM is the standard EM tool GTeX ships. Dropping GReinSS in — with a trivial $\Pr(X|S)$ — matches long-read ground truth far better. Panel c: on MBD2, GReinSS recovers the two true isoforms with near-correct proportions; RSEM splits mass across wrong isoforms. Panel d: distribution of (GReinSS - RSEM) error is shifted negative → GReinSS wins across the genome.

17. GReinSS already powers two cancer methods

GReinSS isn’t just a new paper method — it’s the generalization of machinery that already produced two cancer-genomics tools. If you work on trees,

CNAs, isoforms, or any grow-able discrete structure with a known likelihood, this framework likely applies to you.

18. When should you reach for GReinSS?

Decision guide. The two hard requirements: an incremental generator and a likelihood. If you have those, the four-line recipe is all you need to start — exactly what the notebook demonstrates.

19. Thank you — let's build

Wrap up: the method is one reward formula with a clean theorem, it beats the standard tools on simulations and on real isoform data, and it's a drop-in for discrete latent-state problems in cancer genomics. Open the notebook and try it on your own $\Pr(X|S)$.